- *Next.js*—Provides a framework for building responsive web interfaces using synergizing reasoning and acting in language models (ReAct)
- *Vercel AI SDK*—Responsible for connecting the AI and UI state components and streaming components to the UI
- *LangChain.js*—Powers the backend document-processing pipelines, which enables prompt management and handling of multistep workflows
- *LLMs*—Drive the generative AI capabilities, performing summarization from user input

Figure 7.1 provides a visual representation of the typical workflow for this application.

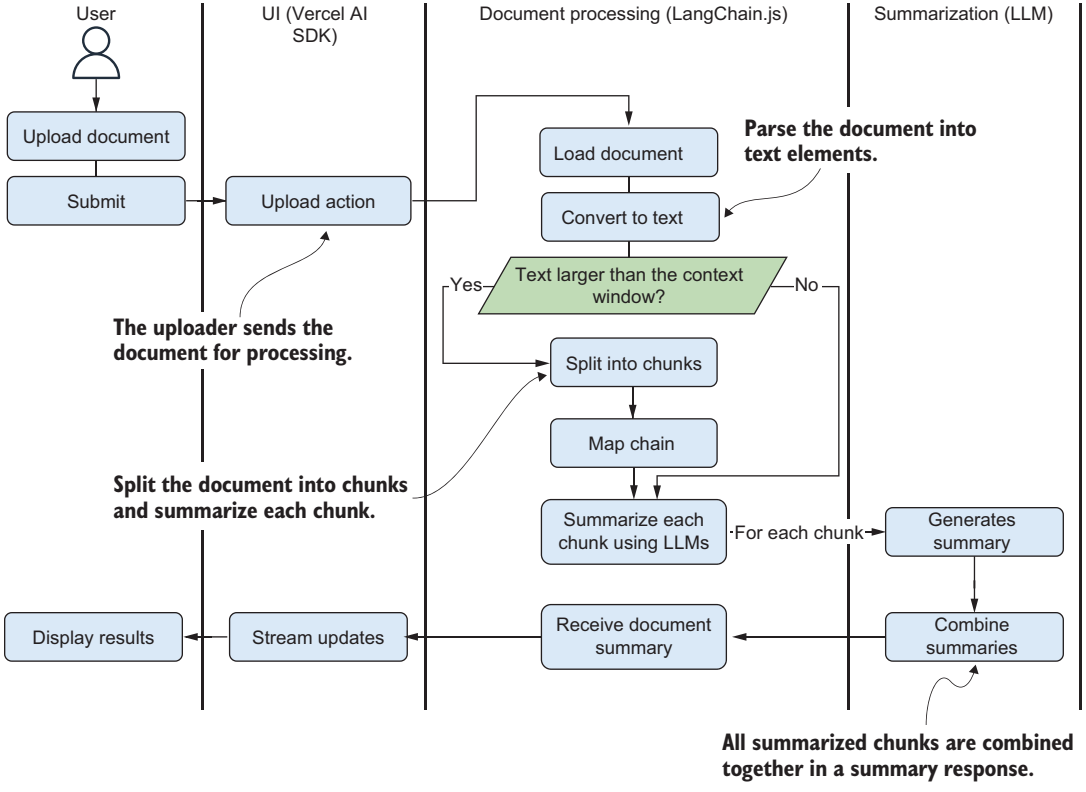


Figure 7.1 The architecture of a document summarization system integrating LangChain.js and the Vercel AI SDK. The user uploads a document, which is processed by LangChain.js; it checks if the document exceeds the context window, splits it into chunks, and summarizes each chunk using LLMs. The individual summaries are then combined by the summarization (LLM) component, which is displayed to the user through the UI.

The summarization step illustrates one approach to processing a large document into manageable chunks using the *MapReduce* method. This technique involves breaking